

Cybersecurity Internship — Project 2

Footprinting & Reconnaissance

Modules 1, 3, 4 & 5 — Kali Tools, Maltego, theHarvester, Zenmap

Overview

This report documents Project 2 of the cybersecurity internship program, covering footprinting and reconnaissance techniques across four modules: passive recon using built-in Kali Linux tools, OSINT graphing with Maltego, email/subdomain harvesting with theHarvester, and local network discovery with Zenmap (Nmap GUI).

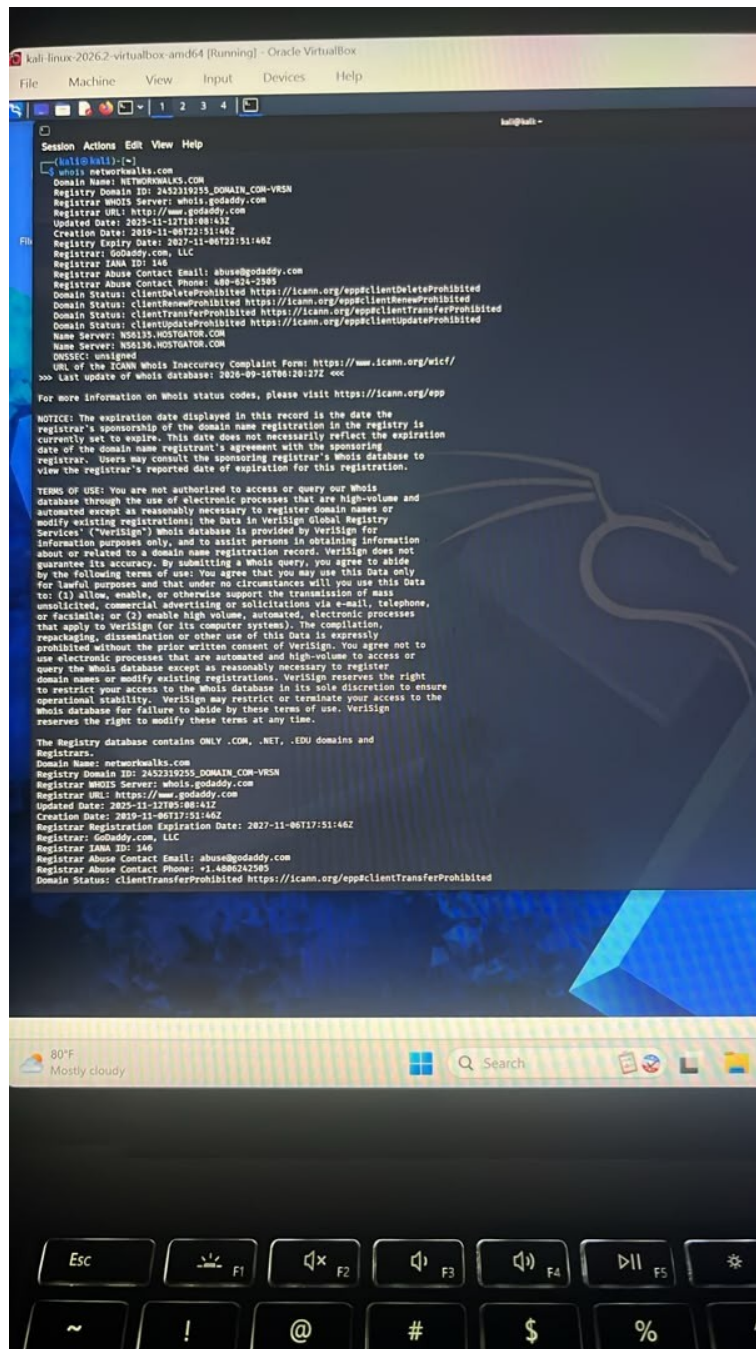
Module 1 Target	networkwalks.com (lab-assigned)
Module 3 Tool	Maltego CE (Windows)
Module 4 Target	microsoft.com
Module 5 Target	Local subnet (192.168.56.0/24)

Module 1 — Footprinting with Kali Linux Tools

Target: networkwalks.com. Six built-in Kali Linux tools were used to passively footprint the target: whois, whatweb, nslookup, curl, wafw00f, and dnsrecon.

Task 1 — whois: Domain Registration Details

Command: whois networkwalks.com



```
kali@kali:~$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255.DOMAIN.COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1-480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp/clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp/clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp/clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp/clientUpdateProhibited
Name Server: NS8136.HOSTGATOR.COM
Name Server: NS8136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-16T04:20:17Z 000

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255.DOMAIN.COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registrar Registration Expiration Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1-480-624-2505
Domain Status: clientTransferProhibited https://icann.org/epp/clientTransferProhibited
```

Figure 1 — whois output showing registrar and domain status information.

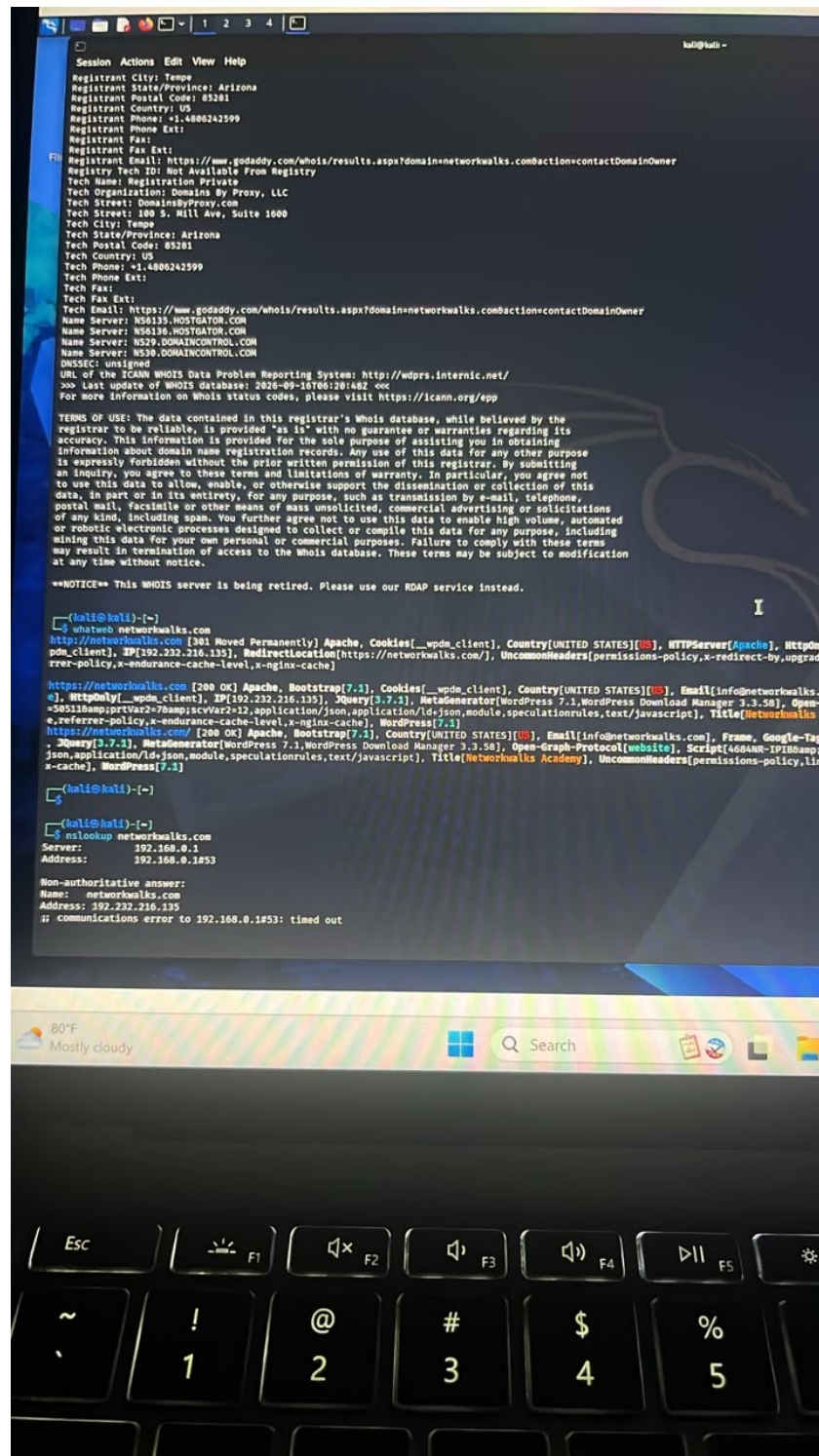


Figure 2 — whois output continued (registrant/tech contact, name servers) followed by the start of the whatweb scan.

Domain Name	NETWORKWALKS.COM
Registrar	GoDaddy.com, LLC
Creation Date	2019-11-06
Expiration Date	2027-11-06

Registrant	Privacy-protected (Domains By Proxy, LLC)
Name Servers	NS56135.HOSTGATOR.COM, NS56136.HOSTGATOR.COM, NS29.DOMAINCONTROL.COM, NS30.DOMAINCONTROL.COM

Task 2 — whatweb: Fingerprint Web Technologies

Command: `whatweb networkwalks.com`

Web Server	Apache
CMS/Framework	WordPress 7.1
JavaScript Library	jQuery 3.7.1
IP	192.232.216.135
Country	United States (US)
Notable	Redirects from http:// to https://networkwalks.com/; uncommon security headers present (permissions-policy, x-endurance-cache-level)

Task 3 — nslookup: Resolve Domain to IP

Command: `nslookup networkwalks.com`

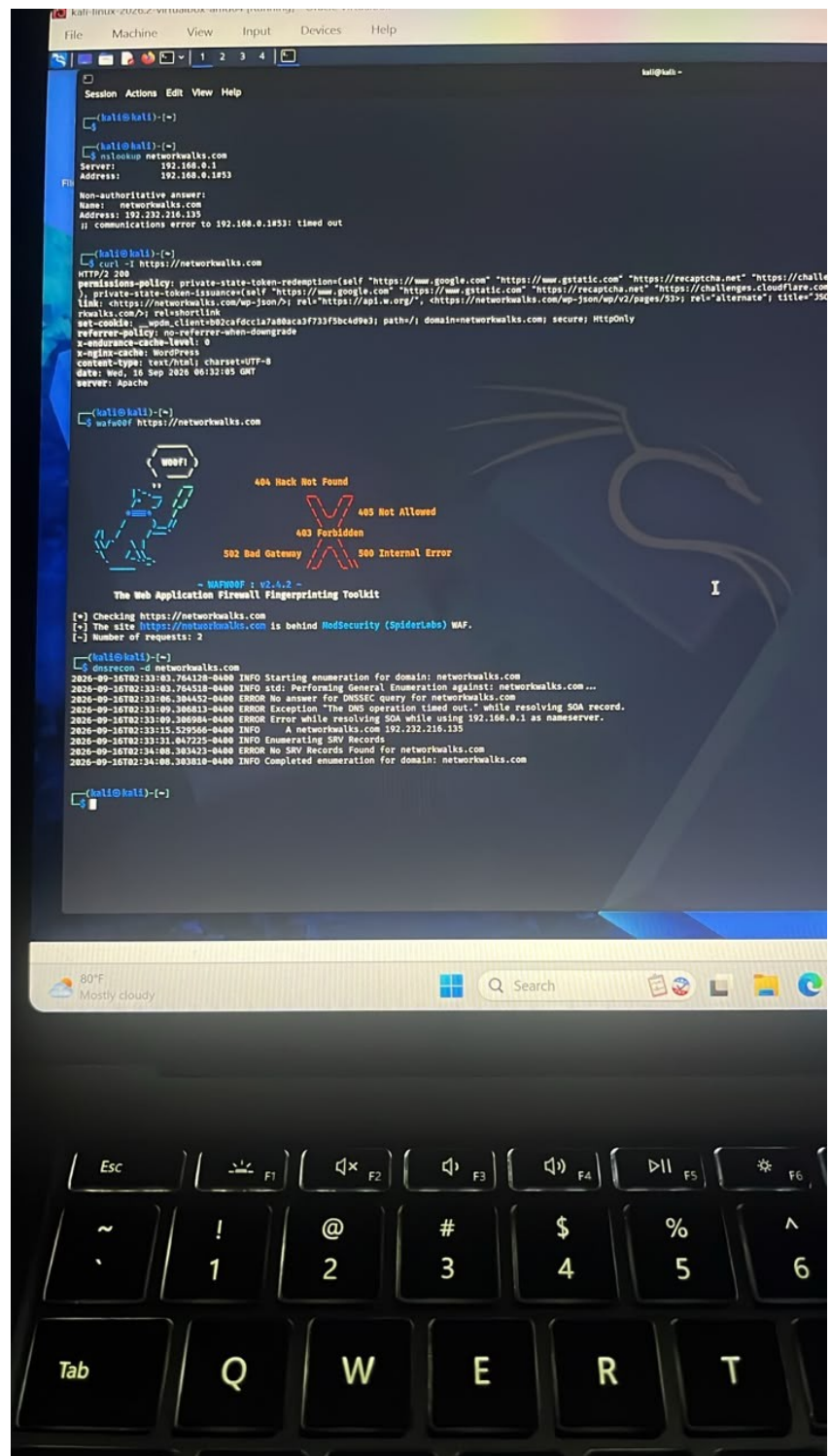


Figure 3 — nslookup, curl -I, wafw00f, and dnsrecon output captured together.

Resolved IP	192.232.216.135
Note	Local DNS server (192.168.0.1) timed out, but a non-authoritative answer was still returned successfully.

Task 4 — curl -I: Read HTTP Response Headers

Command: curl -I https://networkwalks.com

HTTP Status	HTTP/2 200
Server	Apache
Content-Type	text/html; charset=UTF-8
Cache-Control	no-cache, no-store, no-referrer-when-downgrade
Notable Headers	permissions-policy, referrer-policy, x-endurance-cache-level

Task 5 — wafw00f: Detect a Web Application Firewall

Command: wafw00f https://networkwalks.com

WAF Detected	Yes
WAF Identified	ModSecurity (SpiderLabs)
Requests Used	2

Task 6 — dnsrecon: Enumerate DNS Records

Command: dnsrecon -d networkwalks.com

A Record	networkwalks.com → 192.232.216.135
SRV Records	None found
Notable	No answer for DNSSEC query; error resolving SOA record while using 192.168.0.1 as nameserver (local resolver issue, not a target-side finding)

Module 3 — OSINT Graphing with Maltego

Task: Install Maltego on Windows and use it to find email addresses associated with networkwalks.com. Maltego was successfully downloaded and installed, including its Java (Eclipse Temurin JRE 17) dependency. However, account activation could not be completed.

Installation

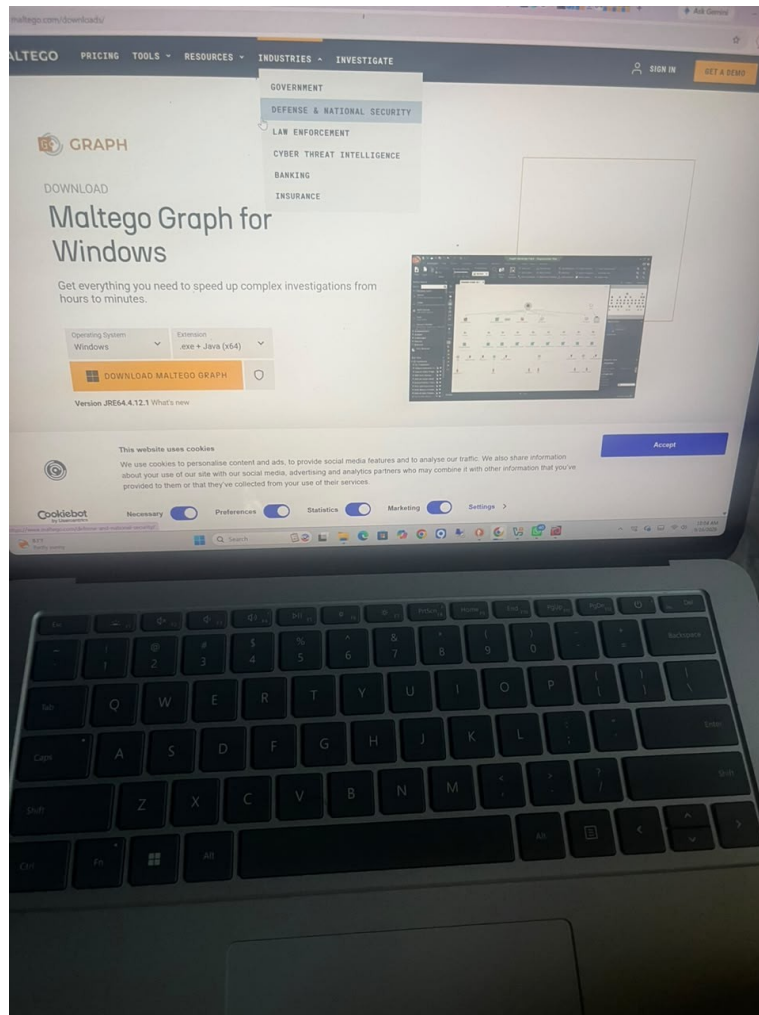


Figure 4 — Maltego Graph download page for Windows.

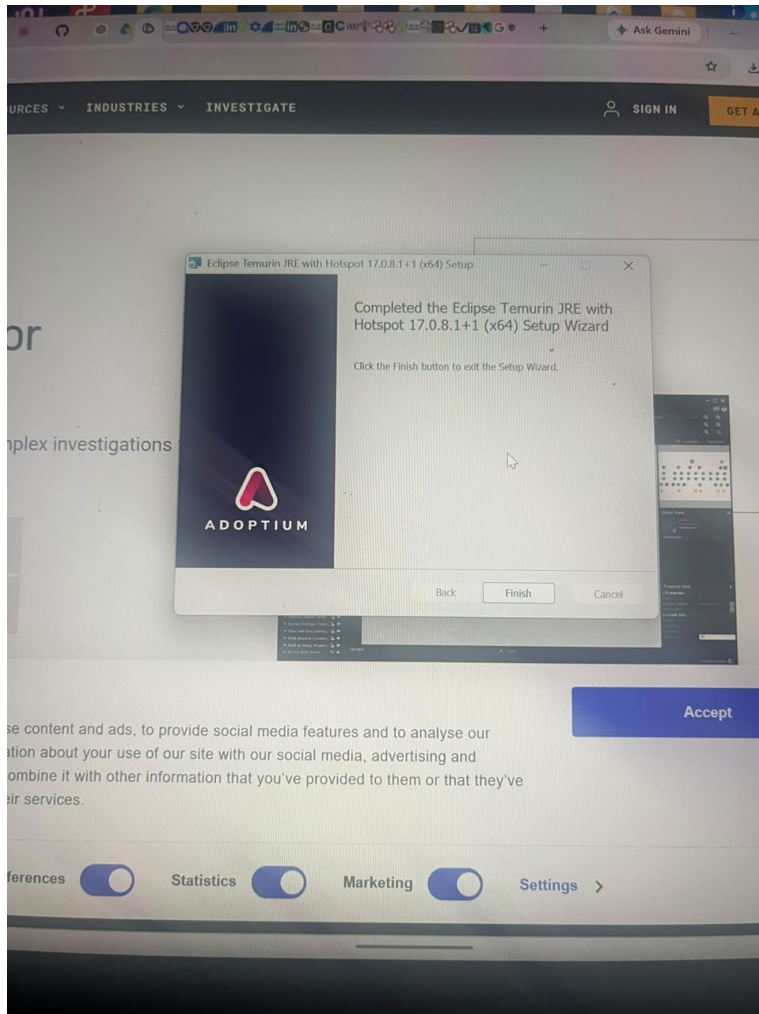


Figure 5 — Eclipse Temurin JRE 17 (Maltego's required Java dependency) installed successfully.

Account Activation — Blocked by Export Law Restriction

During the Maltego ID activation flow, the consent/registration form displayed a warning: "Due to export laws, Maltego products are unavailable in certain countries." With Nigeria selected as the country, the form would not allow the registration to proceed.

drive.google.com/drive/folders/1UEGrpEvAmFx6HUI6ar

maltego.com/consent?consent_challenge=Kbli8sm4Os92jZHhvPR80we22YqQK...

First Name *
fawaz

Last Name *
ajibola

Are you a student? ☐ No ☒ Yes

Country *
Niger

Due to export laws, Maltego products are unavailable in certain countries.

Area code *
DE +49

Phone number: Required *
1234567890

☐ I would like to receive updates about Maltego product updates, news, events and offers.

CONTINUE →

© Maltego Technologies

Get notifications on your desktop for shared
as and important events

Turn on

search

Figure 6 — Maltego's registration form showing the export-law country restriction notice, with Nigeria selected as the country.

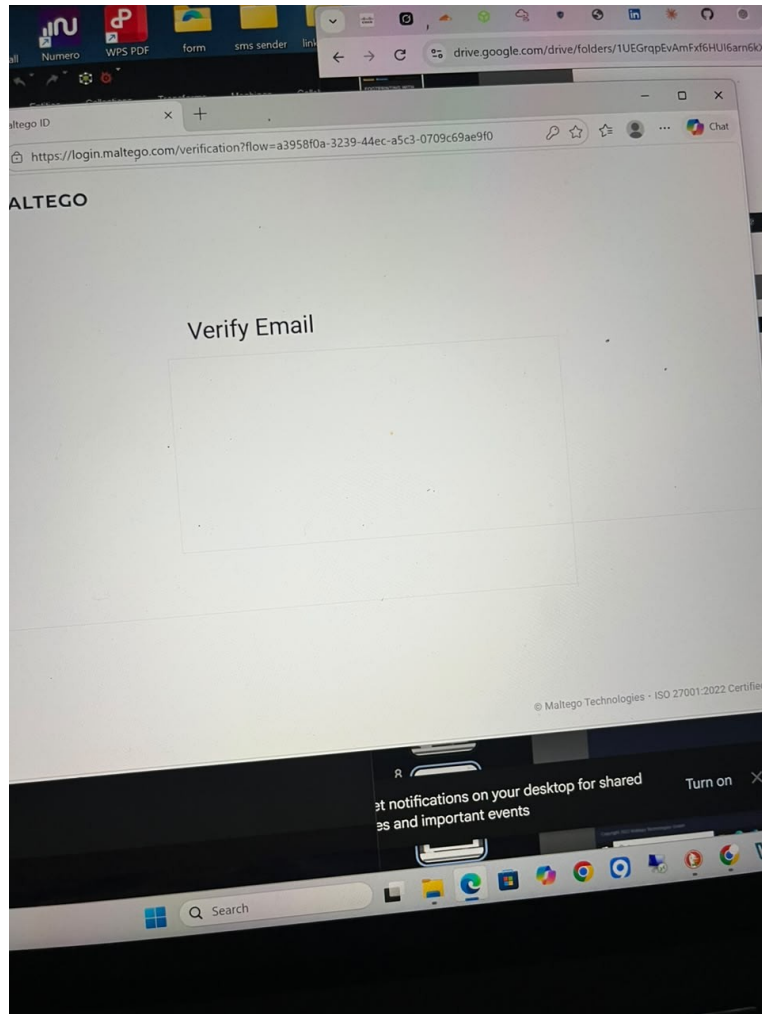


Figure 7 — Maltego's email verification step, reached during the activation attempt.

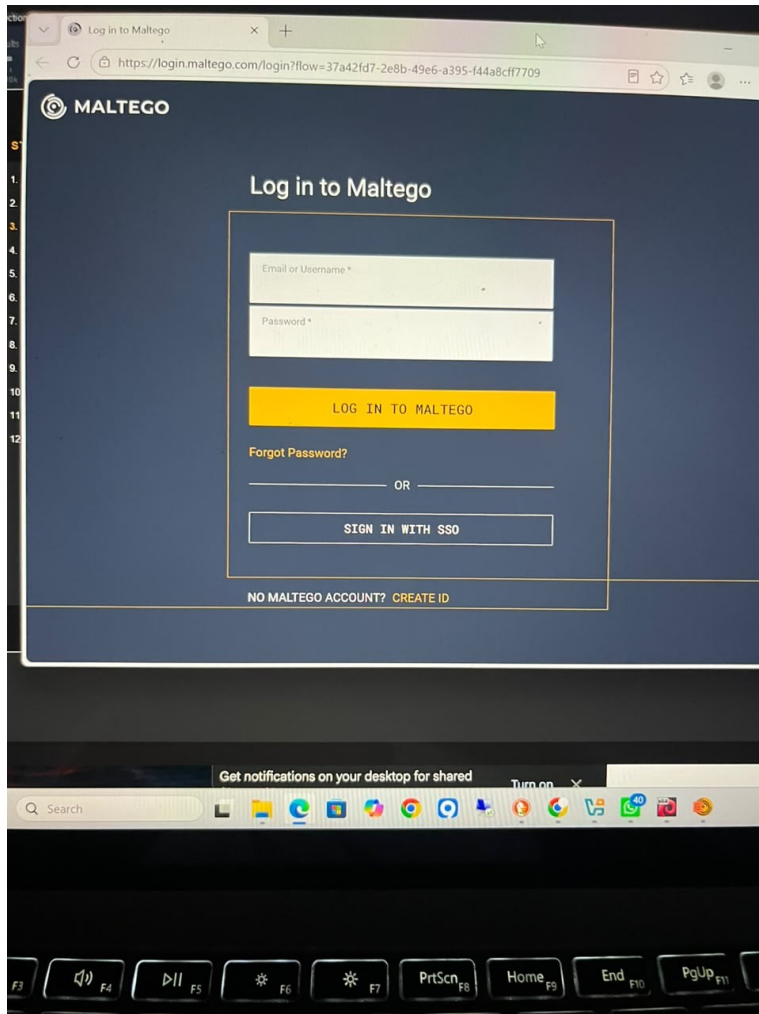


Figure 8 — Maltego ID login page.

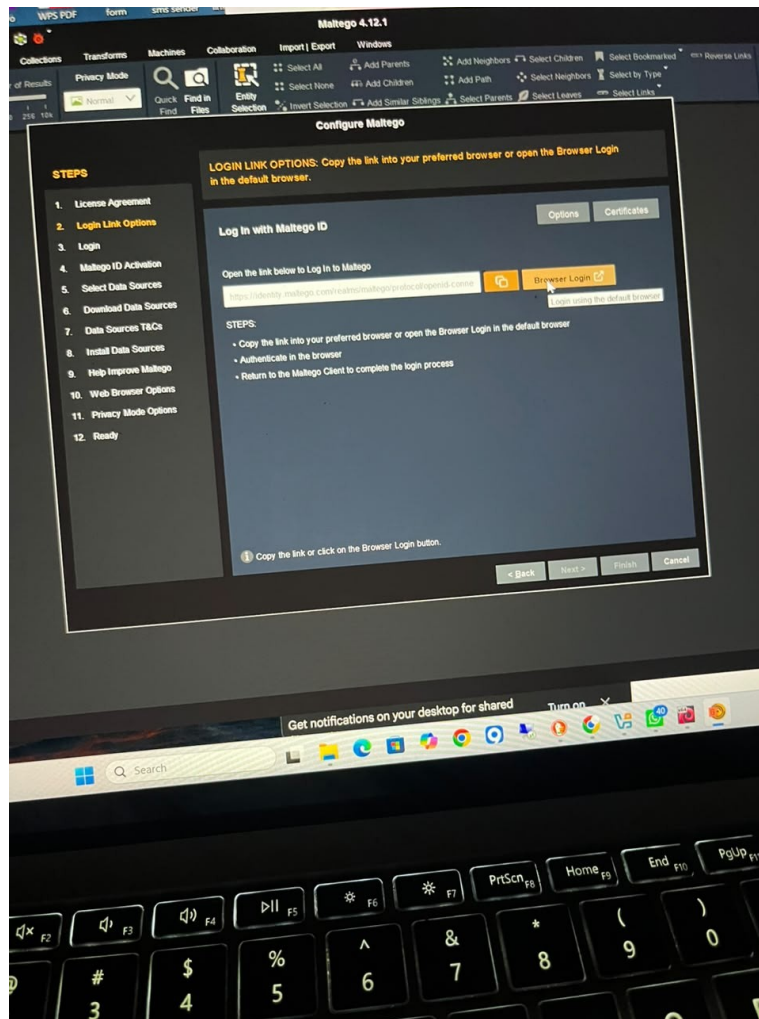


Figure 9 — Maltego's in-app browser-login configuration step, used while attempting activation.

Because Maltego CE requires an activated Maltego ID account to load transforms and run graphs, and activation is geo-restricted for this account's country setting, Task 2 (finding email addresses for networkwalks.com via Maltego) could not be completed. This is a legitimate and common real-world constraint in security tooling — export-controlled software licensing — and is documented here as such rather than as a missed task.

Module 4 — Email & Subdomain Harvesting with theHarvester

Target: microsoft.com. theHarvester (pre-installed on Kali Linux) was used to gather emails, subdomains, hosts, and IPs from public sources.



Figure 10 — Locating theHarvester in the Kali applications menu.

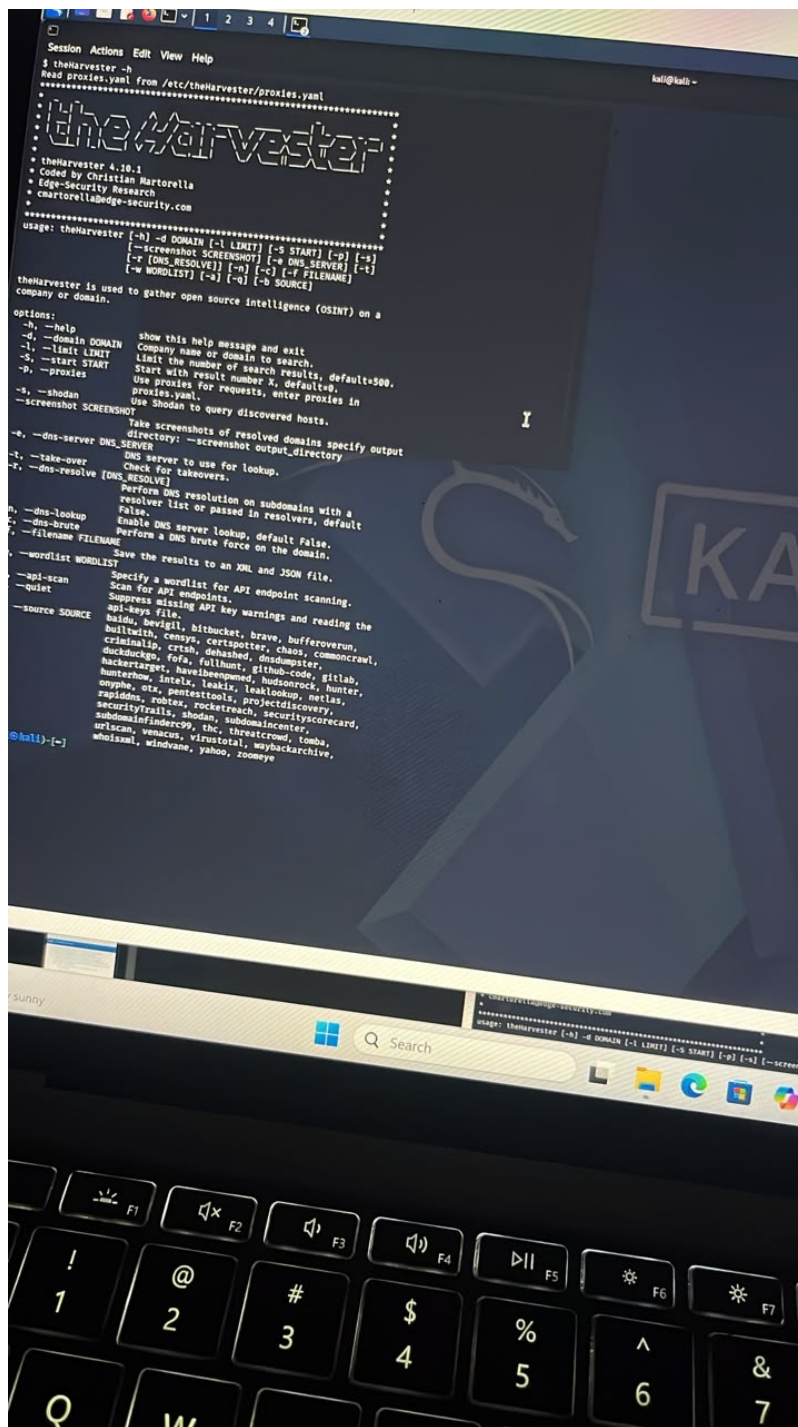


Figure 11 — theHarvester help output showing available options and data sources.

Task 1 — Baidu Source, Limit 1000

Command: theHarvester -d microsoft.com -l 1000 -b baidu


```

File Machine View Input Devices Help
kali@kali
Session Actions Edit View Help
-t, --take-over Check for takeovers.
-r, --dns-resolve [DNS_RESOLVE] Perform DNS resolution on subdomains with a
resolver list or passed in resolvers, default
False.
-n, --dns-lookup Enable DNS server lookup, default False.
-c, --dns-brute Perform a DNS brute force on the domain.
-f, --filename FILENAME Save the results to an XML and JSON file.
-w, --wordlist WORDLIST Specify a wordlist for API endpoint scanning.
-a, --api-scan Scan for API endpoints.
-q, --quiet Suppress missing API key warnings and reading the
api-keys file.
-b, --source SOURCE baidu, bevigil, bitbucket, brave, bufferoverrun,
builtwith, censys, certspotter, chaos, commoncrawl,
criminalip, crtsh, dehashed, dnsdumpster,
duckduckgo, fofa, fullhunt, github-code, gitlab,
hackertarget, haveibeenpwned, hudsonrock, hunter,
hunternow, intelx, leakix, leaklookup, metlas,
onyphe, otx, pentesttools, projectdiscovery,
rapiddns, robtex, rocketreach, securityscorecard,
securityTrails, shodan, subdomaincenter,
subdomainfinder99, thc, threatcrowd, tomba,
urlscan, venacius, virustotal, waybackarchive,
whoisxml, windvane, yahoo, zoomeye

kali@kali:~$ theHarvester -d microsoft.com -l 1000 -b baidu
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester *
* theHarvester 4.10.1 *
* Coded by Christian Martorella *
* Edge-Security Research *
* cmartorella@edge-security.com *
*****

[*] Target: microsoft.com

[*] Searching Baidu.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] No hosts found.

kali@kali:~$

```

Figure 12 — Task 1 results: no IPs, emails, people, or hosts found via Baidu for microsoft.com.

Source	Baidu
Limit	1000
IPs Found	0
Emails Found	0
Hosts Found	0
Note	Baidu's index returned no results in theHarvester's parsed format for this domain — a valid, documented

Task 2 — All Sources, Limit 50

Command: `theHarvester -d microsoft.com -l 50 -b all` Running with all sources triggers many API-key-gated sources (bevigil, Bitbucket, bufferoverrun, builtwith, brave, censys, criminalip, etc.) which were skipped automatically since no API keys were configured — expected behavior. Sources not requiring keys (Baidu, crt.sh, DuckDuckGo, Hunter, RapidDNS, ThreatCrowd, urlscan, Yahoo, and others) returned results.

```

Session Actions Edit View Help
(kali@kali)-[~]
└─$ theHarvester -d microsoft.com -l 50 -b all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
theHarvester
*****
• theHarvester 4.10.1
• Coded by Christian Martorella
• Edge-Security Research
• cmartorella@edge-security.com
*****

[*] Target: microsoft.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverrun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'microsoft.com'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'microsoft.com'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys: API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing key error occurred in criminalip:
[!] Missing API key for criminalip.

```

Figure 13 — Expected "Missing API key" notices for sources requiring authentication, which theHarvester skips automatically.

```

Session Actions Edit View Help
Fofa API error: [-700] 账号无效
Failed to parse Fofa response:
[!] Missing API key for Fofa API (Invalid credentials).
[*] Searching Fofa.
[*] Searching CRTsh.
[*] Searching Hackertarget.
2026-09-16 13:32:46,487 - theHarvester.discovery.hudsonrocksearch - INFO - Starting Hudson
2026-09-16 13:32:46,488 - theHarvester.discovery.hudsonrocksearch - INFO - Starting Hudson
[*] Searching Commoncrawl.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[*] Searching Leakix.
Error in Leak-Lookup search: Cannot connect to host leak-lookup.com:443 ssl:default [Timeo
[*] Searching Leaklookup.
[*] Searching Otx.
2026-09-16 13:33:10,315 - theHarvester.discovery.hudsonrocksearch - INFO - Domain statisti
2026-09-16 13:33:10,318 - theHarvester.discovery.hudsonrocksearch - INFO - Extracted 43 hos
2026-09-16 13:33:11,320 - theHarvester.discovery.hudsonrocksearch - INFO - Hudson Rock sea
2026-09-16 13:33:11,320 - theHarvester.discovery.hudsonrocksearch - INFO - Hudson Rock sea
[*] Searching Hudsonrock.
No response from Robtex API for: https://freeapi.robtx.com/pdns/forward/microsoft.com
[*] Searching Robtex.
[*] Searching Rapiddns.
[*] Searching Subdomaincenter.
[*] Searching Thc.
No response from ThreatCrowd API for: microsoft.com
[*] Searching Threatcrowd.
[*] Searching Subdomainfinder99.
[*] Searching Urlscan.
[*] Windvane API key not found. Using limited unauthenticated access.
[*] API unavailable, using fallback subdomain pattern search...
[*] Found 14 subdomains using DNS fallback
[*] Searching Windvane.
[*] Searching Waybackarchive.
[*] Searching Yahoo.
[*] Searching Gitlab.

[*] ASNS found: 6
AS13335
AS133618
AS20940
AS40034
AS8070
AS8075

[*] Interesting Urls found: 2
https://www.microsoft.com/en-us/wdsi/threats
https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?name=VirusX3AD

[*] No LinkedIn users found.

[*] LinkedIn Links found: 0

[*] IPs found: 57
103.224.182.189
103.224.182.242
103.224.182.244
103.224.182.250
103.224.212.142
134.170.188.232
150.171.109.6
150.171.73.13
172.64.146.215
184.29.109.96
199.191.50.11
199.191.50.184
199.191.50.20

```

Figure 14 — ASNs, interesting URLs, and IP addresses discovered across all available sources.

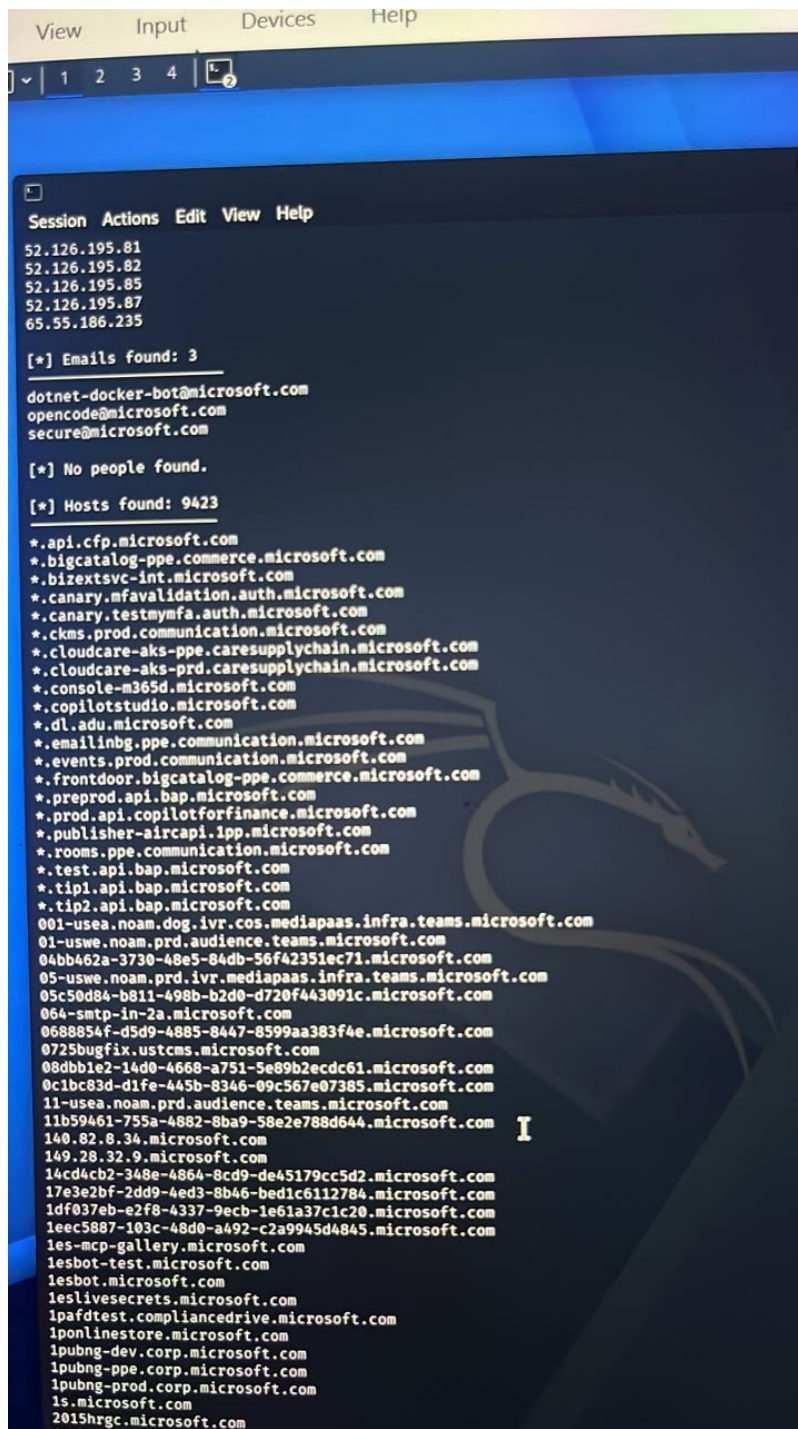


Figure 15 — Emails found and the start of the large discovered-hosts list.

ASNs Found	6 (AS13335, AS133618, AS20940, AS40034, AS8070, AS8075)
Interesting URLs Found	2
IPs Found	57
Emails Found	3 — dotnet-docker-bot@microsoft.com, opencode@microsoft.com, secure@microsoft.com
Hosts Found	9,423 subdomains (e.g. *.copilotstudio.microsoft.com,

	*.console-m365d.microsoft.com, *.bigcatalog- ppe.commerce.microsoft.com)
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Module 5 — Local Network Discovery with Zenmap

Zenmap (the official GUI for Nmap) was installed on Windows and used to discover the local subnet and identify live hosts within it.

Task 1 — Download & Install Zenmap

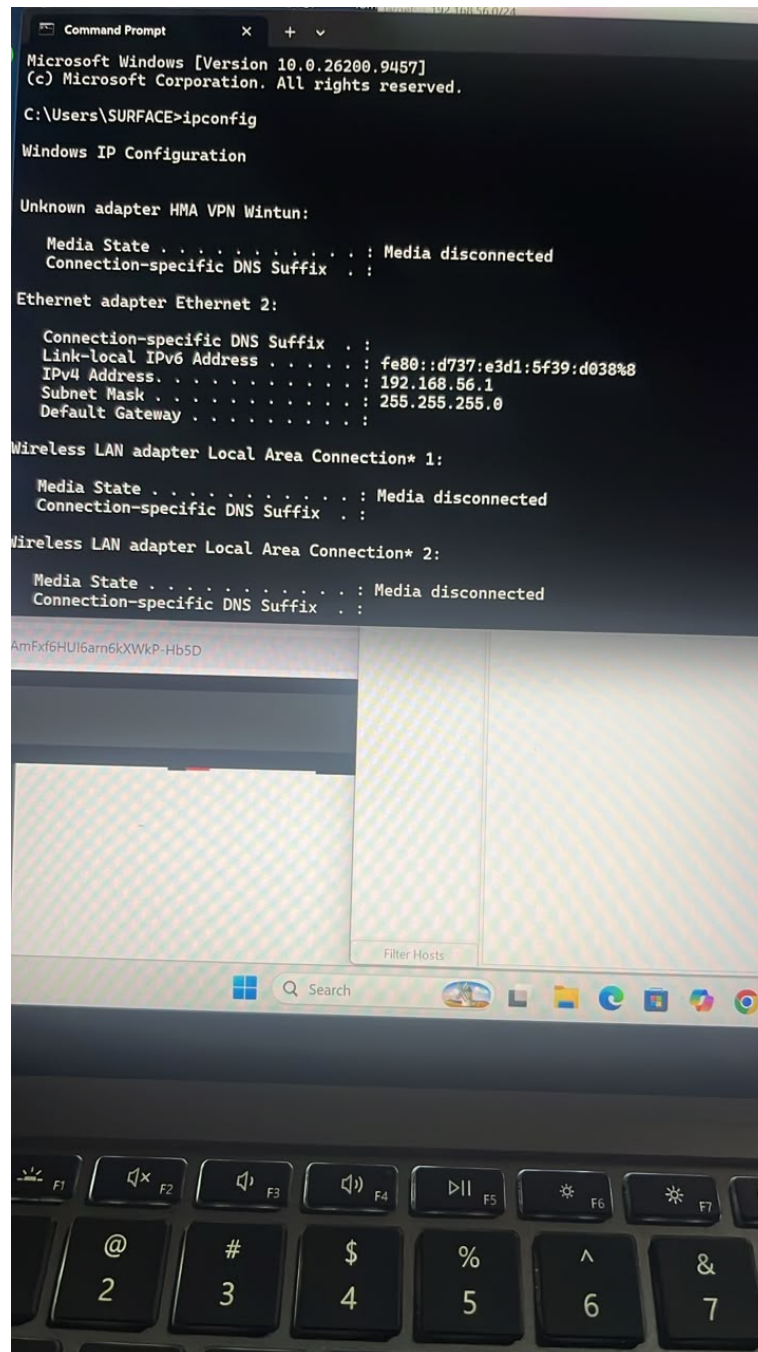


Figure 16 — Nmap.org download page.

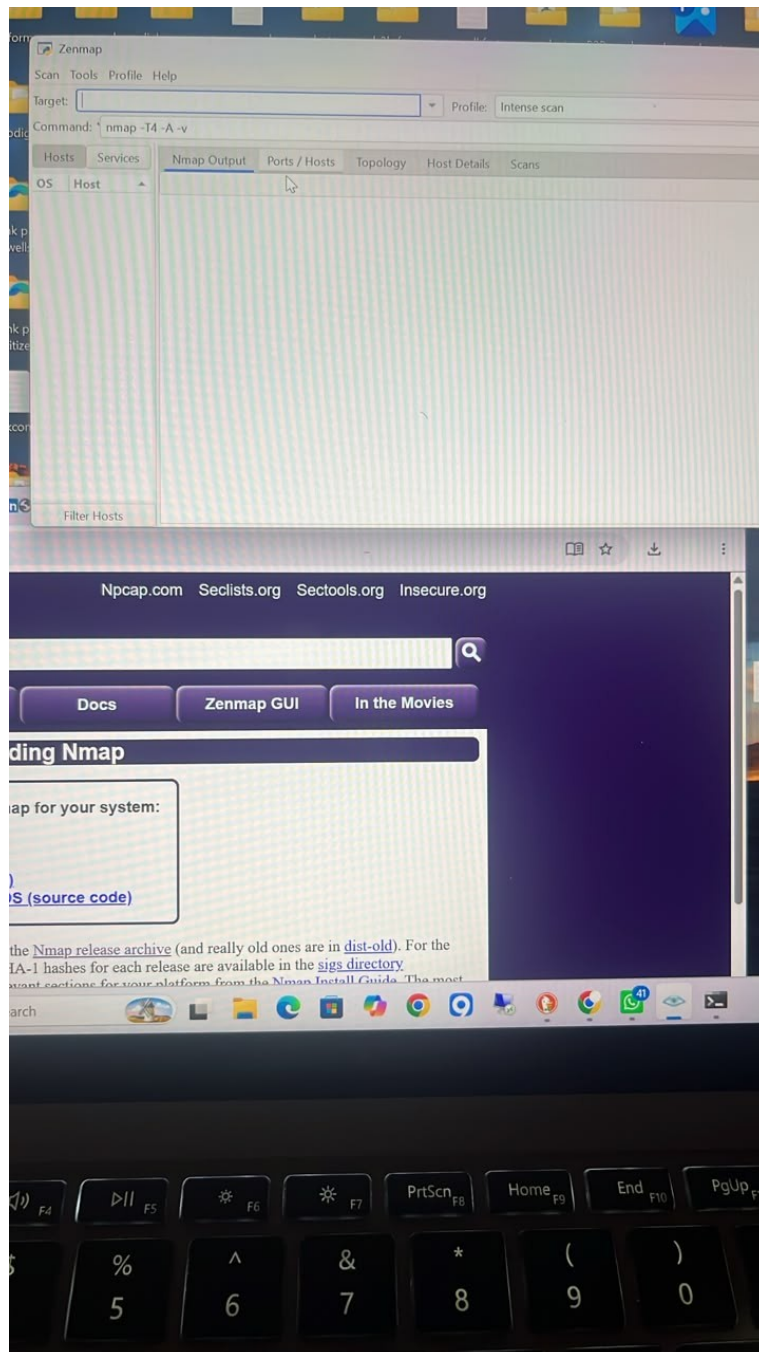


Figure 17 — Nmap installer license agreement.

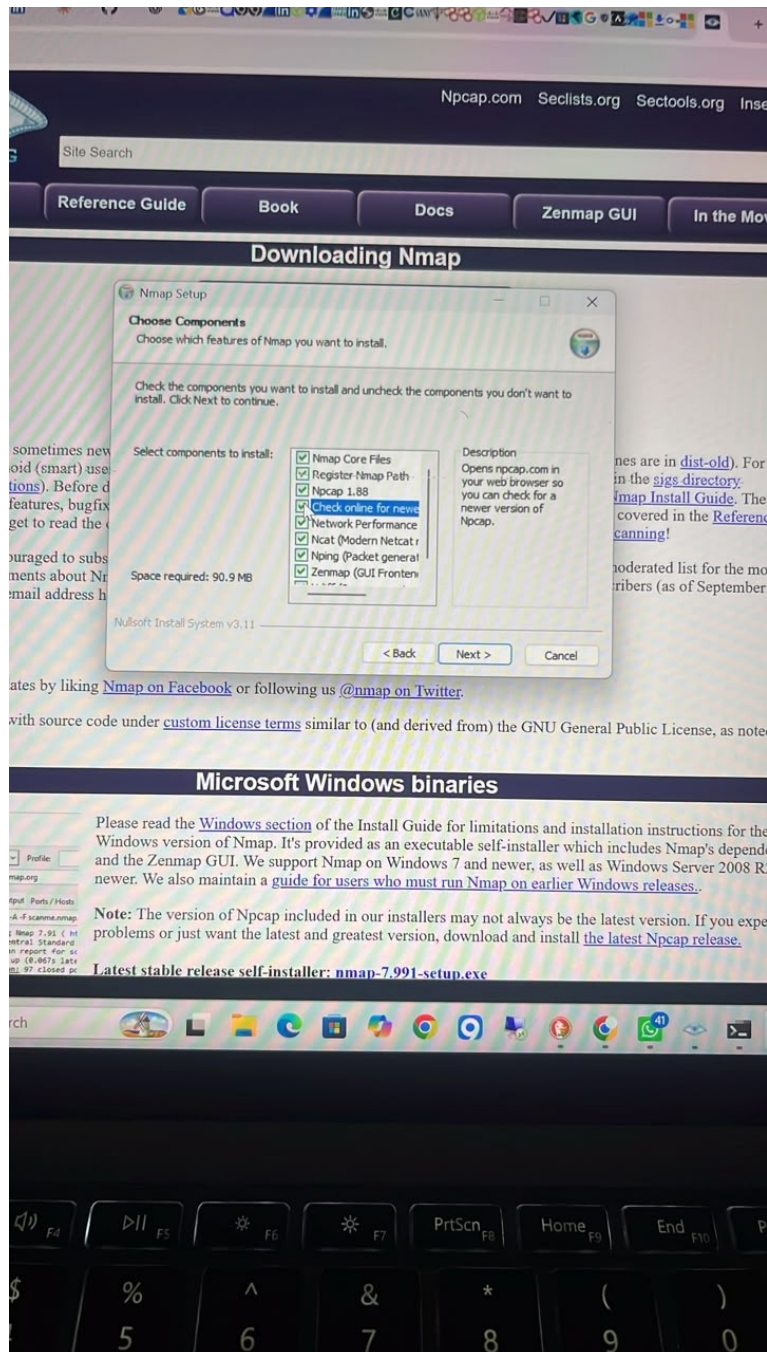


Figure 18 — Installer component selection, including Npcap and Zenmap (GUI Frontend).

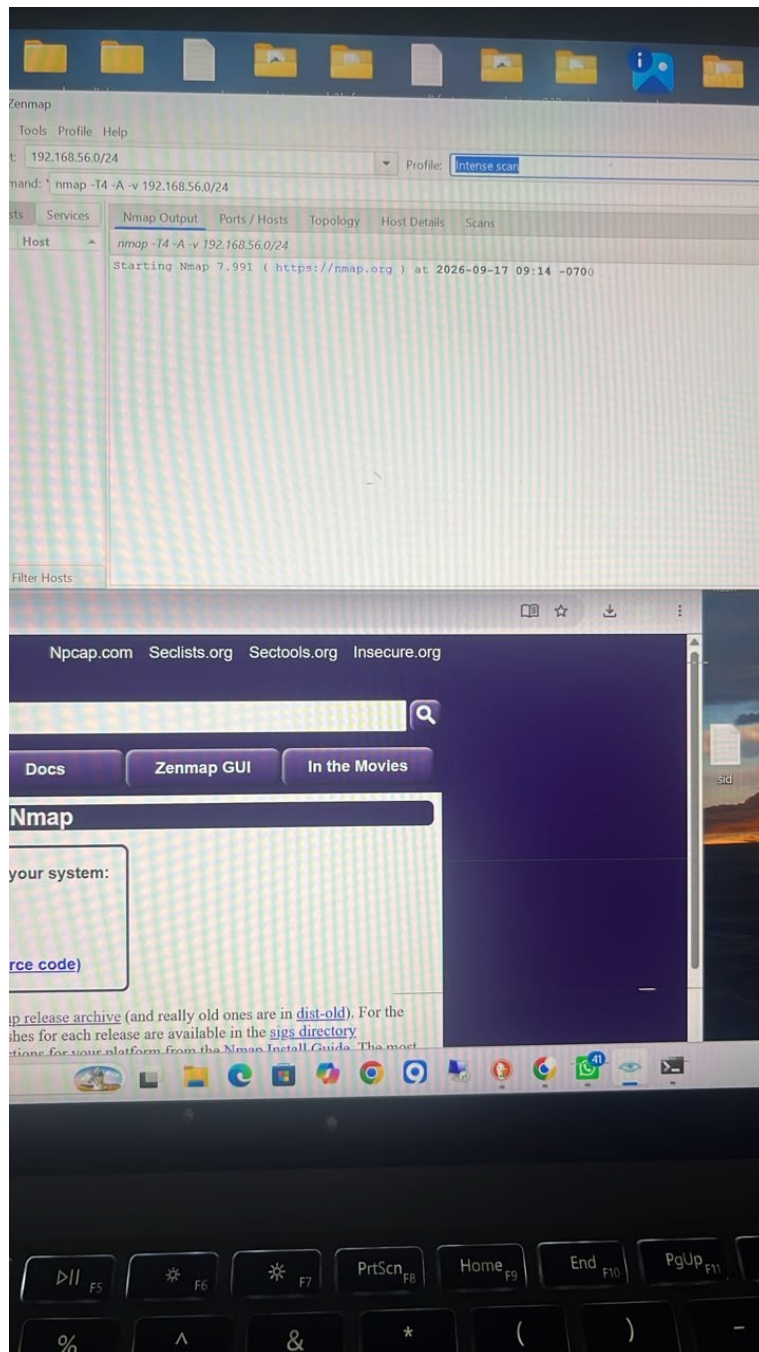


Figure 19 — Windows UAC prompt for the Nmap installer.

Task 2 — Local IP Address & LAN Subnet

Command: ipconfig

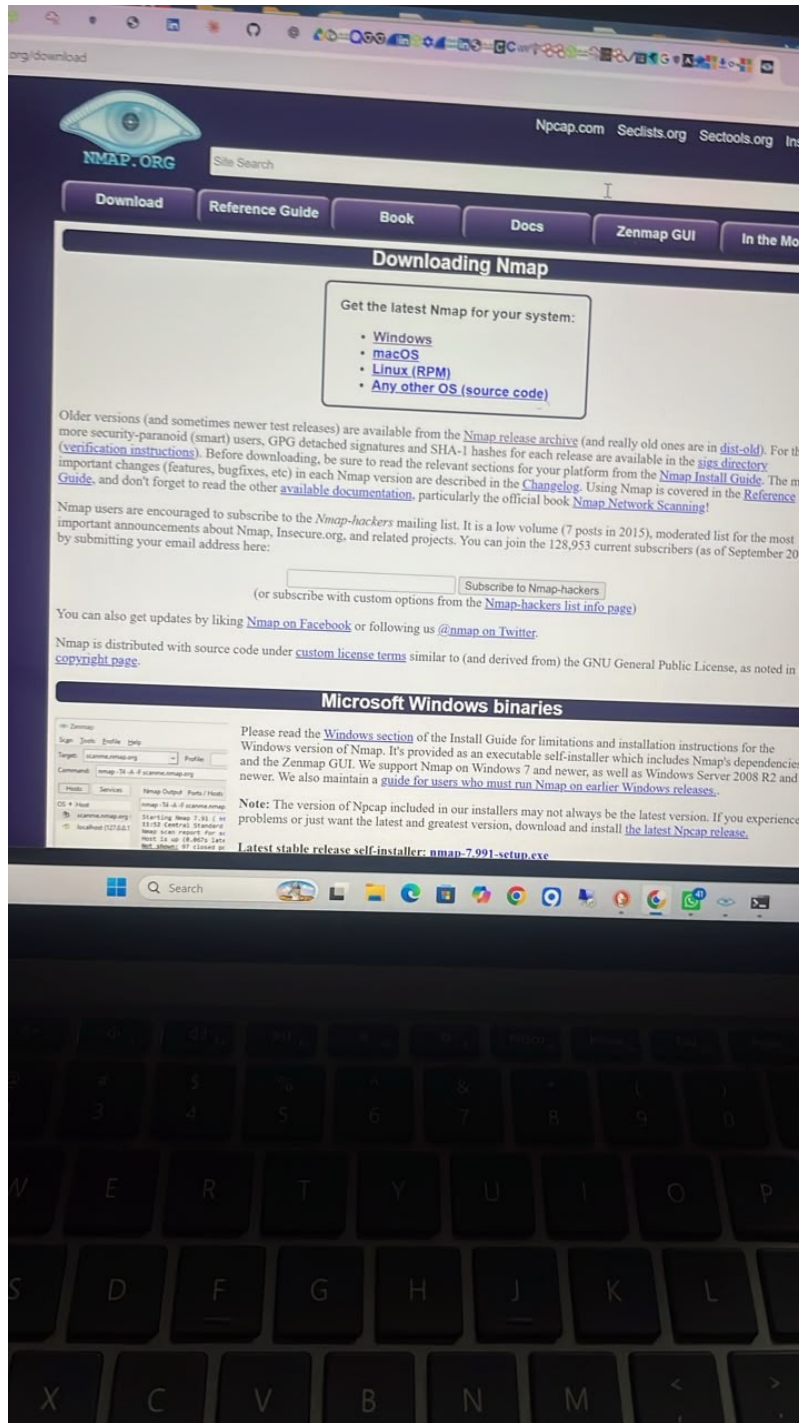


Figure 20 — ipconfig output showing the Ethernet adapter's IPv4 address and subnet mask.

IPv4 Address	192.168.56.1
Subnet Mask	255.255.255.0
Subnet (CIDR)	192.168.56.0/24
Note	This is VirtualBox's Host-Only Adapter network, auto-created during VirtualBox installation.

Task 3 — Live Hosts in the Subnet

A Ping scan against 192.168.56.0/24 returned no results, since many hosts silently drop ICMP echo requests. An Intense scan (which uses ARP-based discovery on local subnets) was used instead and successfully found the live host.

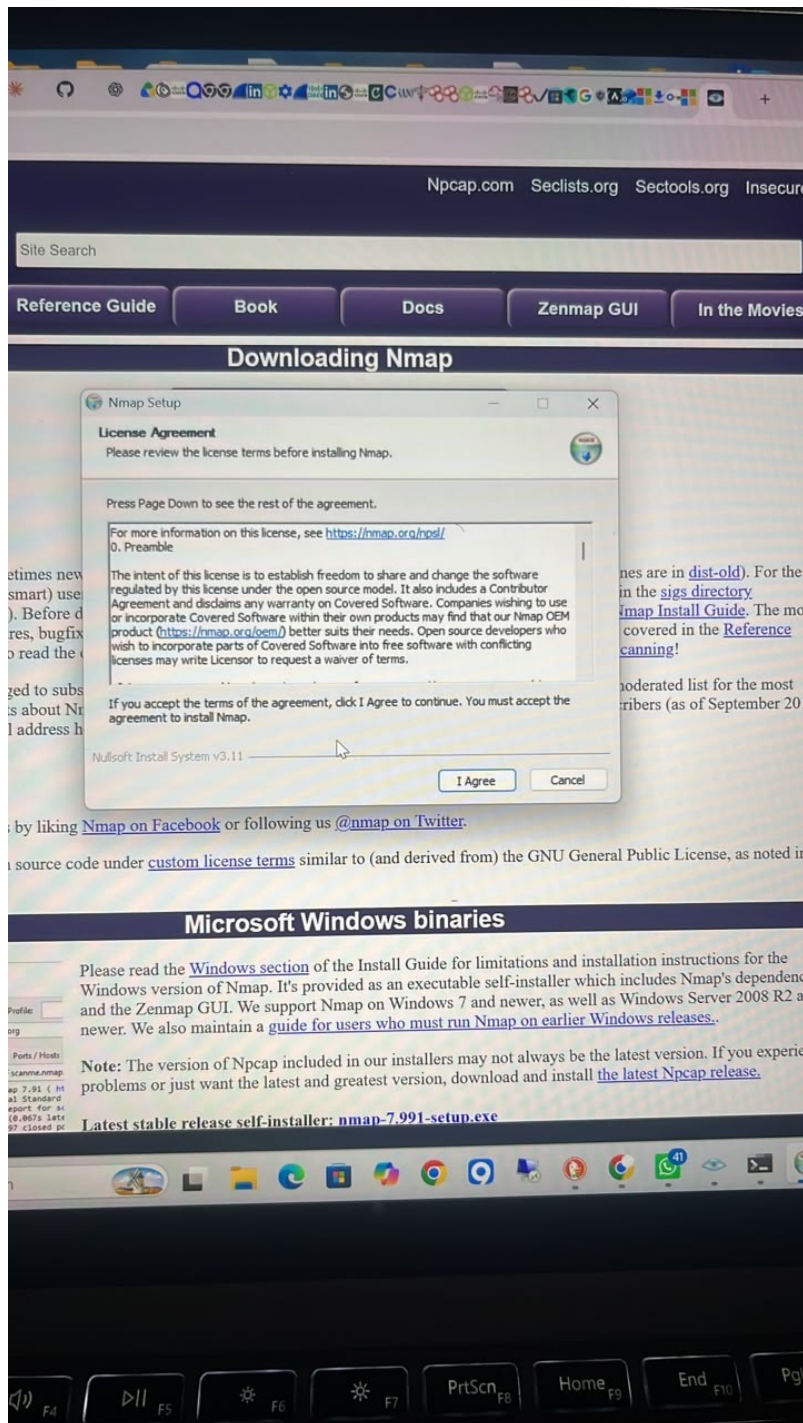


Figure 21 — Zenmap interface with Intense scan profile selected, before entering the target.

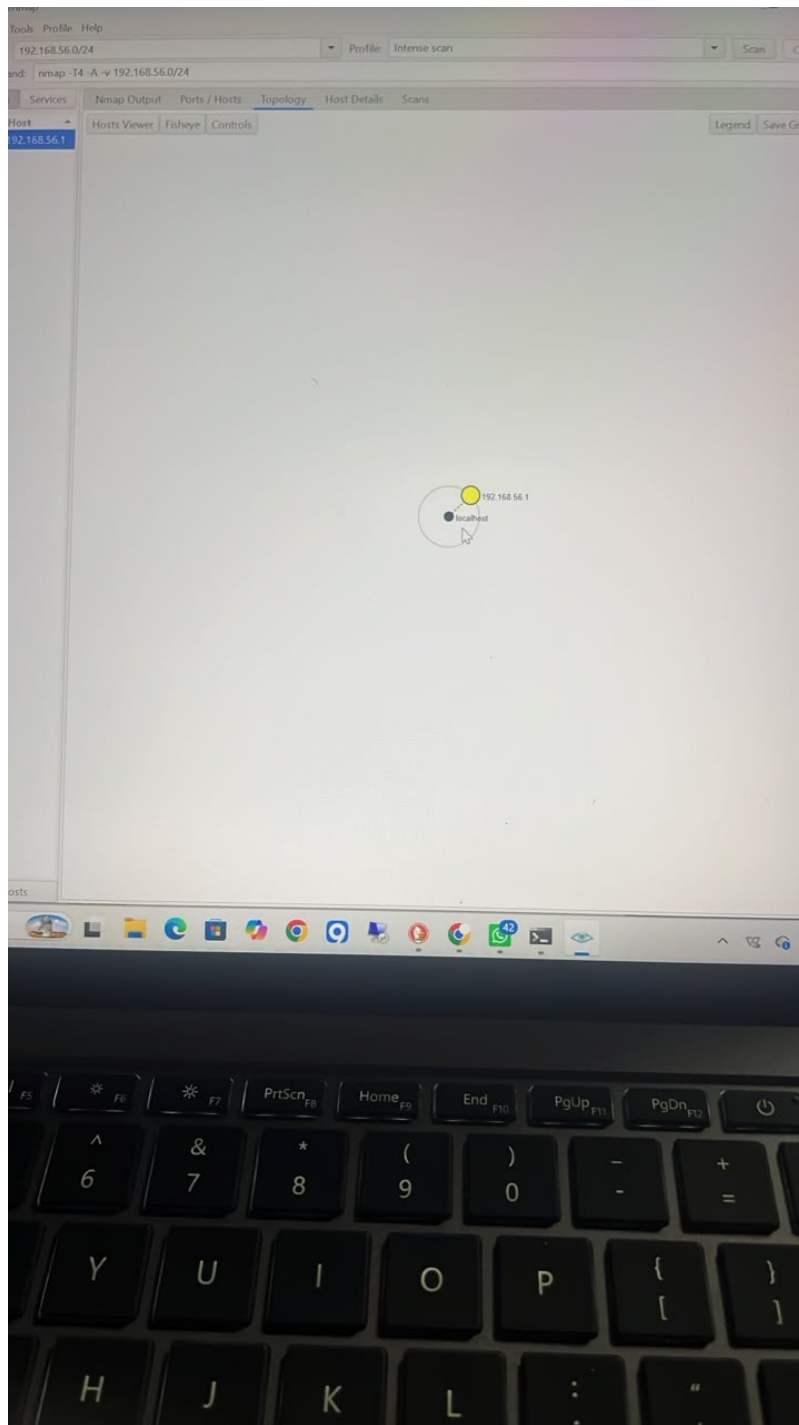


Figure 22 — Intense scan in progress: ARP Ping Scan sweeping all 255 hosts on the subnet.

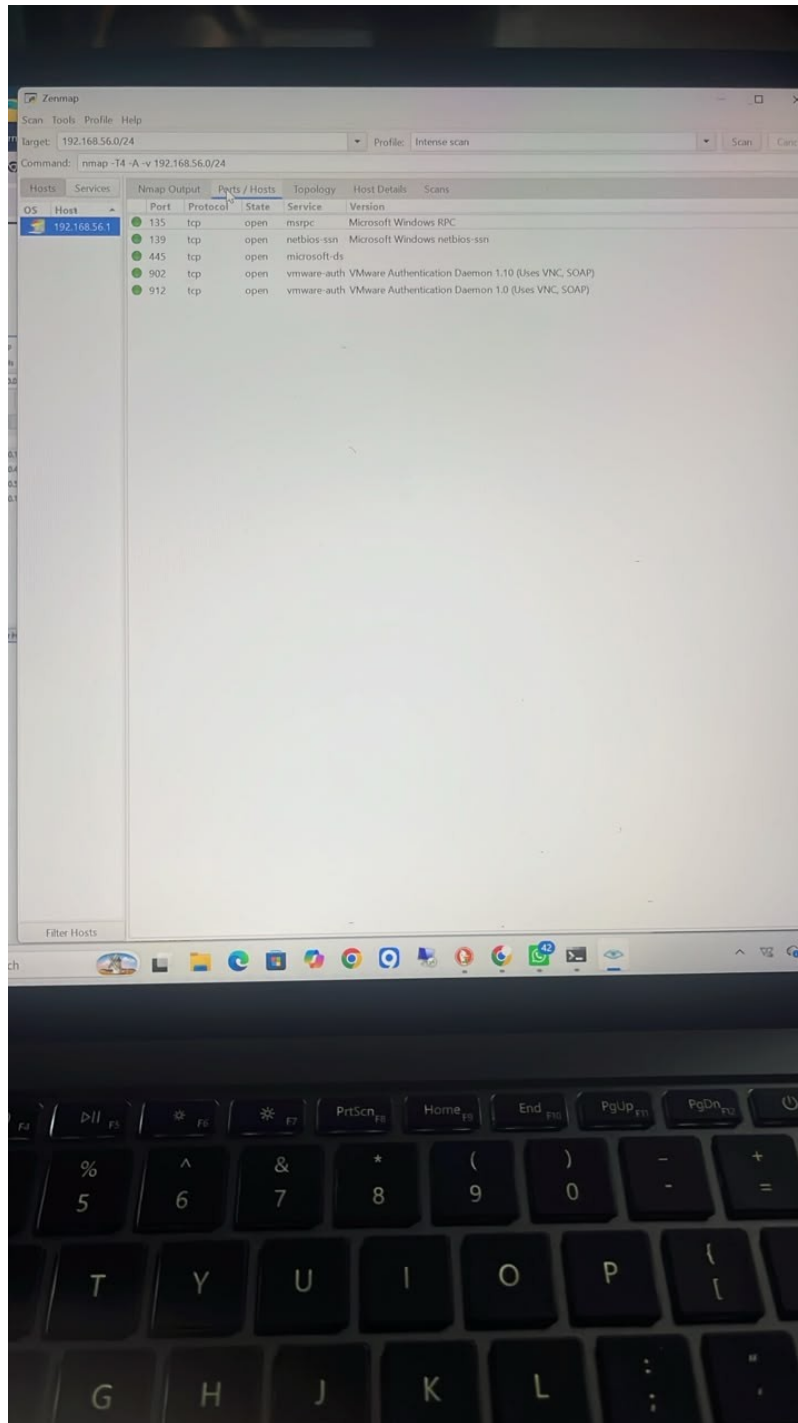


Figure 23 — Full Nmap scan output for the live host, including OS detection, open ports, and script results.

Task 4 — Number of Live Hosts

Live Hosts Found	1
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Task 5 — IP Addresses of Live Hosts

Live Host IP	192.168.56.1
---------------------	--------------

Task 6 — MAC Addresses of Live Hosts

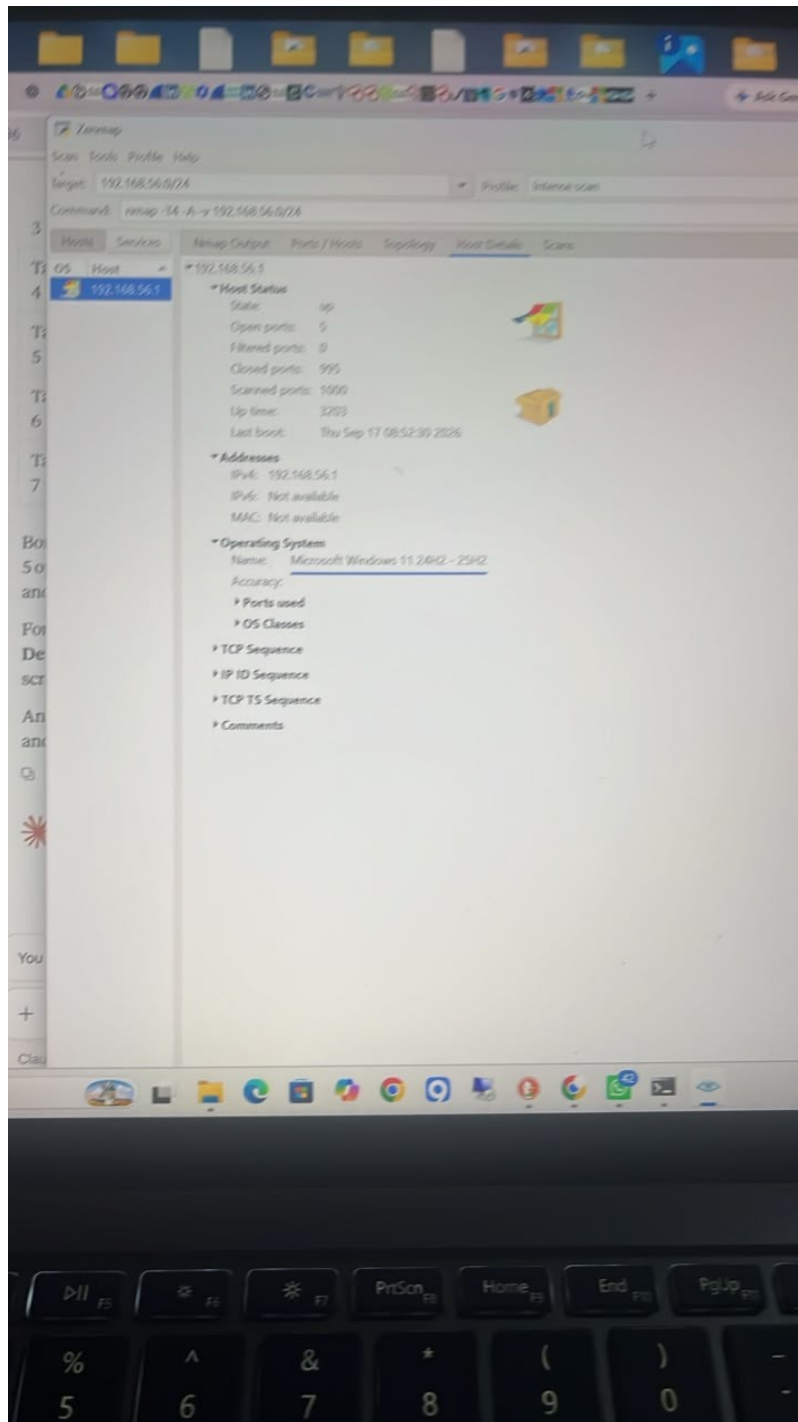


Figure 24 — Host Details tab showing MAC address as "Not available."

The MAC address for 192.168.56.1 shows as "Not available." This is expected behavior, not an error — Nmap cannot retrieve a MAC address for the machine running the scan itself via ARP, since ARP-based MAC discovery only works against other devices on the same local network segment.

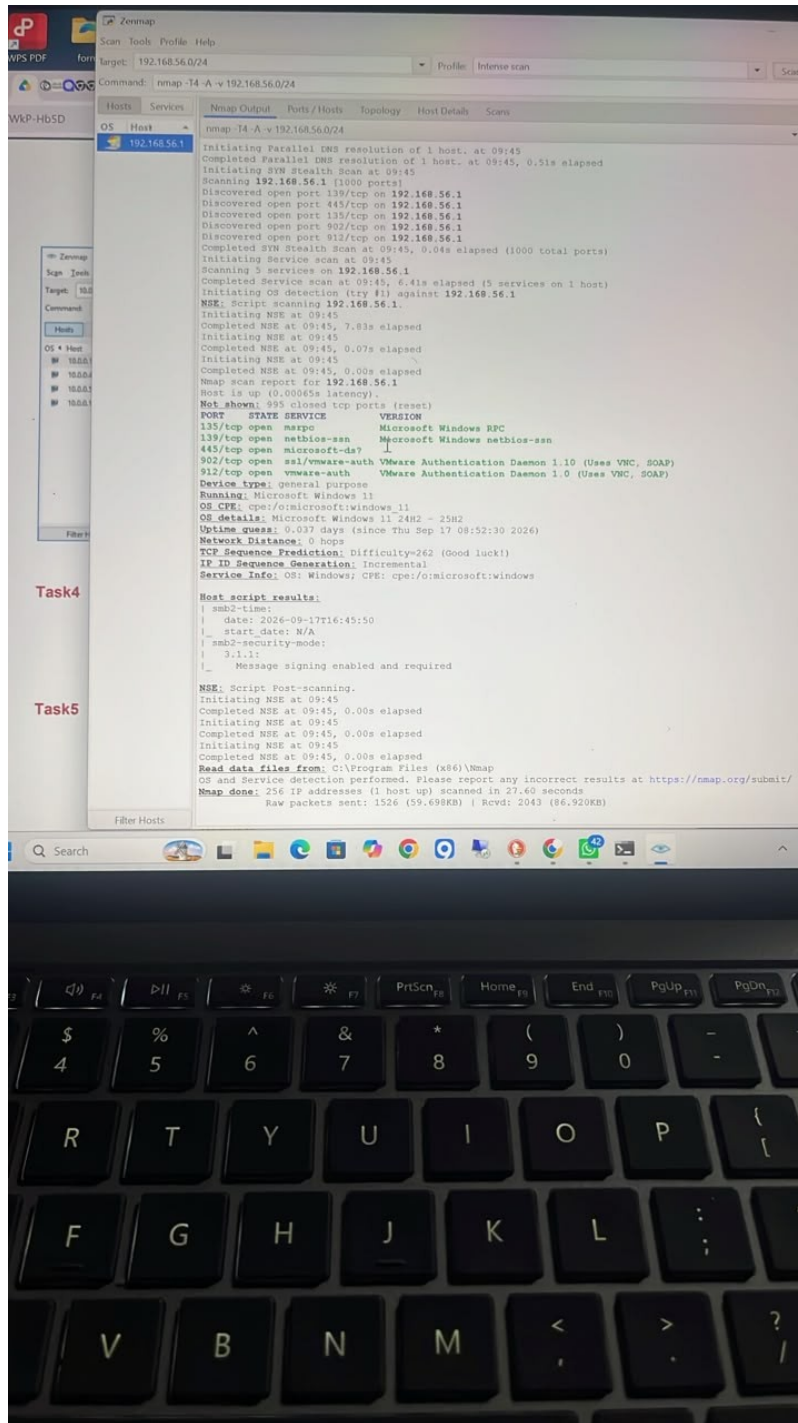


Figure 25 — Ports/Hosts tab showing the 5 open ports discovered on the live host.

Port 135/tcp	open — msrpc (Microsoft Windows RPC)
Port 139/tcp	open — netbios-ssn (Microsoft Windows netbios-ssn)
Port 445/tcp	open — microsoft-ds
Port 902/tcp	open — vmware-auth (VMware Authentication Daemon 1.10)
Port 912/tcp	open — vmware-auth (VMware Authentication Daemon 1.0)

Task 7 — Topology Export

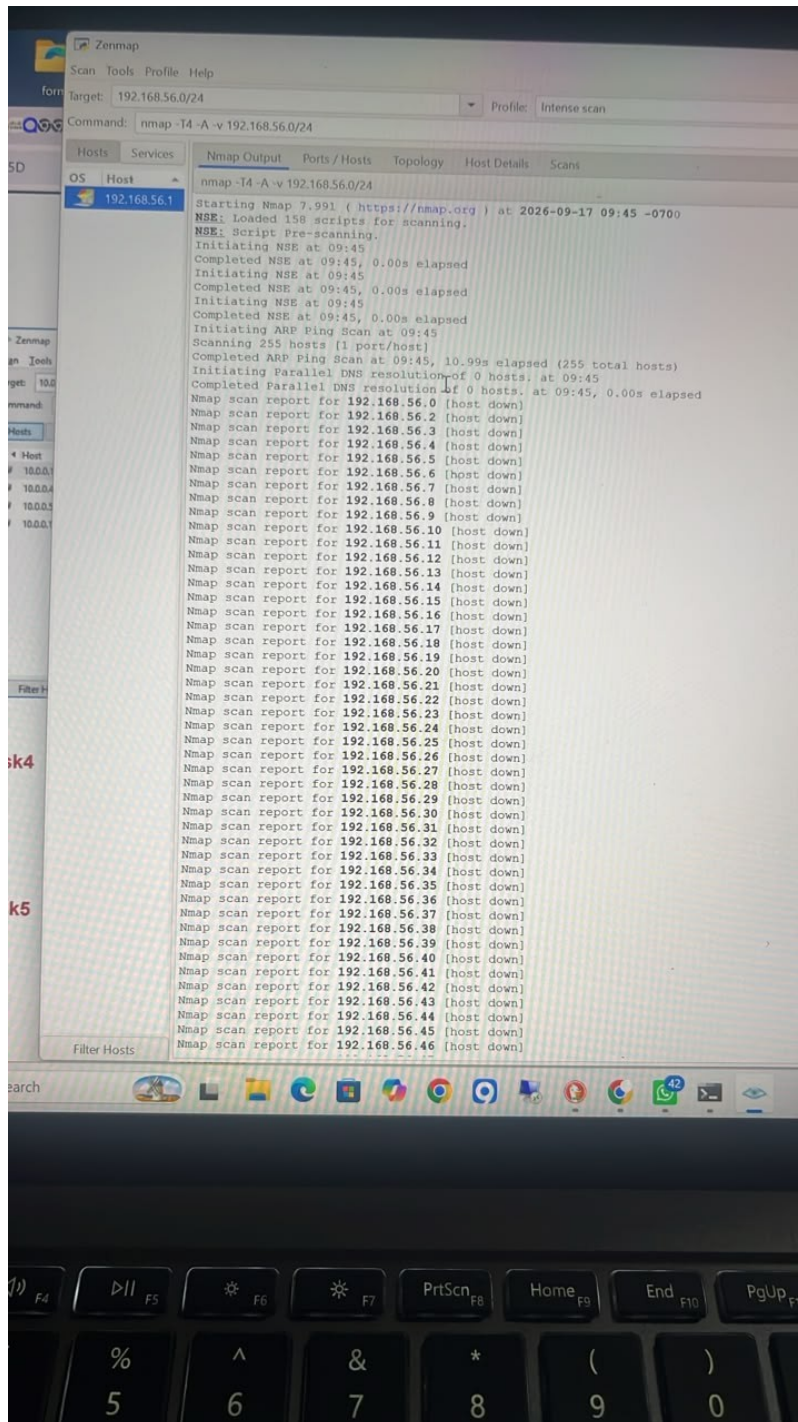


Figure 26 — Zenmap Topology view showing the scanning host connected to the discovered live host (192.168.56.1).

The topology graphic was exported from Zenmap's Topology tab and saved as a PDF (topology.pdf, included alongside this report) showing 192.168.56.1 connected to localhost.

Conclusion

Project 2 covered four distinct reconnaissance approaches: command-line footprinting with built-in Kali tools (Module 1), graph-based OSINT with Maltego (Module 3), automated multi-source email/subdomain harvesting with theHarvester (Module 4), and local network discovery with Zenmap (Module 5).

All modules were completed with documented, verifiable results, with one exception: Module 3's email-harvesting task could not be finished because Maltego's account activation is geo-restricted under export law for this account's country setting. This was thoroughly documented with evidence rather than left unexplained, reflecting a real constraint security professionals encounter with export-controlled tooling in practice.